Subqueries

Agenda

1. Concept
2. Types of subquery
3. Using IN, ANY and ALL
4. Create, insert, delete and update in subquery
5. Correlated subquery

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1. Concept

- What is subquery
- Need
- Syntax

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2. Types of subquery

- Types
  1. Single-row subqueries
  2. Multiple-row subqueries
  3. Using IN, ANY and ALL

3. Create, insert, delete and update in subquery

OBJECT TECHNOLOGIES

1. **Create in subquery**

```
sql>create table table_name
      as
     (
      select * from table  ;
     );
sql>create table table_name
      as
     (
      select * from table
      where 1=2  ;
     )
```

2. **Insert in subquery**

```
sql>insert into empTemp
     (
      select empno,ename,sal
      from emp
      where deptno =20
     );
```

3. **Delete in subquery**

```
sql>delete
     from emp11
     where deptno=(select deptno
                   from (select deptno
                         from emp11
                         where
                         ename='Smith')
                   tempTable);
```

4. **Update in subquery**

```
sql>update emp
     set sal = select sal
               from (select sal
                     from emp
                     where ename='Black')
                     tempTable)
               where ename = 'Ward';
```

4. Correlated Query

- Simple correlated query

```
mysql>select *
        from emp e1
        where sal > (select avg(sal)
                     from emp e2
                     where e1. deptno =
e2.deptno);
```

-inner query executes first.

-output of inner query will be compared with the row fetched by outer query.

-bottom up approach.

- Exists and not exists operators

```
mysql>select *
        from emp e
        where exists(select deptno
                     from dept d
            where e.deptno =d.deptno );
```

```
mysql>select *
        from emp e
        where not exists(select deptno
                     from dept d
            where e.deptno =d.deptno );
```